

JEE MAIN REVISION MASTER: CURRENT ELECTRICITY SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 52 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kirchhoff's Laws, Wheatstone Bridge & Potentiometer	35 PYQs	DC Instruments
Very High	Ohm's Law, Drift Velocity & Resistance Temperature	22 PYQs	Conduction

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Kirchhoff's Laws, Wheatstone Bridge & Potentiometer	$\frac{P}{Q} = \frac{R}{S}$ (Wheatstone), $\frac{E_1}{E_2} = \frac{I_1}{I_2}$, $r = R \left(\frac{I_1}{I_2} - 1 \right)$, $S = \frac{I_g}{I - I_g} R$, $R_{\text{series}} = \frac{V}{I_g} - G$
Ohm's Law, Drift Velocity & Resistance Temperature	$I = n e A v_d$, $v_d = \frac{e E \tau}{m}$, $\rho = \frac{m}{n e^2 \tau}$, $R(T) = R_0(1 + \alpha \Delta T)$, $\vec{J} = \sigma \vec{E}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kirchhoff's Laws, Wheatstone Bridge & Potentiometer (Very High Yield — 35 PYQs)

Core Principle: Kirchhoff's junction and loop rules, balanced bridges, potentiometer EMF comparison, and galvanometer conversion.

■ **Caution / JEE Pitfall:** A potentiometer measures EMF with zero drawn current, making it infinitely superior to a real voltmeter which always draws finite current.

■ **Speed Trick:** To convert galvanometer to ammeter: connect small shunt S in parallel. To voltmeter: connect high resistance R in series.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Refer to the figure given below. The values of I_1, I_2 and I_3 are ____" [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "Two resistors of $200\ \Omega$ and $400\ \Omega$ are connected in series with a battery of 100 V . A bulb rated at 200 V , 100 W is connected ac..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "In the potentiometer, when the cell in the secondary circuit is shunted with $4\ \Omega$ resistance, the balance is obtained at the length 120 cm ..." [Ans: A]

3.2 Ohm's Law, Drift Velocity & Resistance Temperature (Very High Yield — 22 PYQs)

Core Principle: Microscopic model of electrical conduction, electron mobility $\mu = v_d/E$, and thermal variation of resistivity.

■ **Caution / JEE Pitfall:** For conductors, $\alpha > 0$ (resistance increases with temperature). For semiconductors, $\alpha < 0$ (resistance decreases with temperature).

■ **Speed Trick:** When a wire is stretched to n times its initial length while preserving volume, its resistance becomes $n^2 R$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "In the potentiometer, when the cell in the secondary circuit is shunted with 4Ω resistance, the balance is obtained at the length 120..." **[Ans: A]**
- **[2026 Evening Shift] Q#1 (MCQ):** "A battery with EMF E and internal resistance r is connected across a resistance R . The power consumption in R will be maximum when:..." **[Ans: D]**
- **[2026 Morning Shift] Q#2 (MCQ):** "Refer to the figure given below, current between terminals A and B is ____ A...." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Refer to the figure given below. The values of I_1, I_2 and I_3 are ____ .

- (A) $I_1=2.5A, I_2=1.875A, I_3=1.875A$ **[Correct]**
- (B) $I_1=1.875A, I_2=2.5A, I_3=1.875A$
- (C) $I_1=1.875A, I_2=1.875A, I_3=2.5A$
- (D) $I_1=2.5A, I_2=2.5A, I_3=1.875A$

Correct Answer: A

Solution Analysis: Using Kirchhoff's current law, the net accumulation of current at a junction is zero. So, at junction A, $I_1 - I_3 - I_{AD} = 0 \Rightarrow I_{AD} = I_1 - I_3$ And at junction D, $I_{AD} + I_2 - I_{DB} = 0 \Rightarrow I_1 - I_3 + I_2 - I_{DB} = 0 \Rightarrow I_{DB} = I_1 - I_3 + I_2$ Applying Kirchhoff's voltage law in closed loop I, $-4(I_1 - I_3) - 2(I_1 + I_2 - I_3) - I_1 + 10 = 0$
 $-7I_1 - 2I_2 + 6I_3 = -10$... (i) In loop (II) $I_1 + 10 - 4I_3 - 2(I_3 - I_2) = 0$ $2I_2 - 6I_3 - I_1 = -10$... (ii) In loop III, $-5 - 2(I_3 - I_2) + 2(I_1 + I_2 - I_3) = 0$ $2I_1 + 4I_2 - 4I_3 = 5$ (iii) From e

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Morning Shift (Q#1)

In the potentiometer, when the cell in the secondary circuit is shunted with 4Ω resistance, the balance is obtained at the length 120 cm of wire. Now when the same cell is shunted with 12Ω resistance, the balance is shifted to a length of 180 cm . The internal resistance of cell is ____ Ω

- (A) 4 **[Correct]**
- (B) 12
- (C) 6
- (D) 3

Correct Answer: A

Solution Analysis: Let the current in the primary circuit is i_p . If the resistance of potentiometer wire of length L is R , then the resistance of the balancing length l will be $R_l = ((R / L)l)$ The voltage drop at point A is.
 $V = i_p R_l = (\frac{i_p R l}{L}) = k \cdot l$, where k is constant. the balancing length l is proportional to the terminal potential difference V : $V = k \cdot l$... (i) When a cell of EMF E and internal resistance r is shunted (connected in parallel) with an external resistance R , The current in the secondary circuit is, $i_s = (E / R + r)$ Then termina